Rebuttal to Reviewer FNYZ

Additional Baselines. To further demonstrate the robustness of VaSST, we performed comparisons with PySR (Cranmer, 2023), Operon (Burlacu et al., 2020), and DSR (Petersen et al., 2019) in addition to the competing symbolic regression methods considered in the original submission. Here, we summarize the results corresponding to the Feynman equations $\Delta U = Gm_1m_2(1/r_2 - 1/r_1)$ and $P = \kappa A(T_2 - T_1)/d$ as well as a newly added Feynman equation $f = \beta(1 + \alpha \cos\theta)$. For all methods, we use $n = 2000$ (sub-sampled) observations, and reported the out-of-sample RMSEs using the learned symbolic expression on the 10% held-out test set. Different noise regimes have been considered, i.e., from noiseless to noise $\sigma \in (0.1, 0.2)$.

VaSST was run with $K = 3$ trees of depth $D = 3$, using the operator set $\mathcal{O} = \{+, \times, -, /, \exp, \log, \sin, \cos, {}^2\}$. During training, for VaSST we consider reporting the top learned symbolic expression ranked according to the in-sample RMSE. The other hyperparameters are configured the same as in Appendix H.1 of the original submission.

PySR (https://github.com/MilesCranmer/pysr) was run with 200 iterations, 10 populations, population size 50, and maximum expression size 20. The operator set was configured to be the same as that of VaSST.

Operon (C++ implementation https://github.com/heal-research/operon) was run using its symbolic regressor with allowed symbols $\{+, -, \times, /, \sin, \cos, \exp, \texttt{pow}, \texttt{square}, \texttt{constant}, \texttt{variable}\}$. Specifically, we used `generations` = 2000, `population_size=1000`, `max_length` = 20, and `n_threads` = 1.

DSR was run using the `DeepSymbolicRegressor` (https://github.com/dso-org/deep-symbolic-optimization). The operator set used was the same as that for VaSST. The training budget was set to `n_samples` = 200000 candidate expressions with `batch_size` = 500. The policy maximum length was set to `max_length` = 20.

Results. Across the three Feynman equations considered, VaSST consistently recovers the dominant physical structure, with RMSE generally matching the noise level added to the data. For the trigonometric benchmark $f = \beta(1 + \alpha \cos\theta)$, VaSST shows a modest degradation in predictive accuracy, but still achieves strong structural recovery by identifying the core component $\beta(1 + \alpha \cos\theta)$. PySR recovers the exact symbolic form in most cases, except for $f = \beta(1 + \alpha \cos\theta)$, where it returns a more complex surrogate expression. Operon often attains competitive RMSE, but its learned expressions are typically less interpretable and substantially more complex. DSR successfully recovers the true symbolic expression across all cases, with RMSE close to the corresponding noise level. Detailed results are tabulated below.

Results for learning $\Delta U = Gm_1m_2(1/r_2 - 1/r_1)$.

Method	Learned expression	RMSE
VaSST	$\Delta U = -0.000714 + 0.000914\,\frac{G^2 m_1^3 m_2}{r_1^2} - 0.996541\,\frac{Gm_1m_2}{r_1} + 1.010681\,\frac{Gm_1m_2}{r_2}$	0.003656
PySR	$\Delta U = \frac{Gm_1m_2}{r_2} - \frac{Gm_1m_2}{r_1}$	8.61×10^{-16}
Operon	$\Delta U = -0.998001\,\frac{Gm_1m_2}{r_1} + 1.000232\,\frac{Gm_1m_2}{r_2}$	1.34×10^{-5}
DSR	$\Delta U = \frac{Gm_1m_2}{r_2} - \frac{Gm_1m_2}{r_1}$	8.70×10^{-16}

Table 1: Learned symbolic expressions for Feynman equation $\Delta U = Gm_1m_2(1/r_2 - 1/r_1)$ in the noiseless setting from VaSST, PySR, Operon, and DSR along with their RMSEs.

Method	Learned expression	RMSE
VaSST	$\Delta U = -0.003874 + 0.000609\,\frac{G^2 m_1^3 m_2}{r_1^2} - 0.995991\,\frac{Gm_1m_2}{r_1} + 1.017123\,\frac{Gm_1m_2}{r_2}$	0.100194
PySR	$\Delta U = \frac{Gm_1m_2}{r_2} - \frac{Gm_1m_2}{r_1}$	0.100224
Operon	$\Delta U = -0.998605\,\frac{Gm_1m_2}{r_1} + 4.985798\,\frac{\frac{Gm_1m_2}{r_1}}{\sqrt{1 + \left(4.921\,\frac{r_2}{r_1}\right)^2}}$	0.100157
DSR	$\Delta U = \frac{Gm_1m_2}{r_2} - \frac{Gm_1m_2}{r_1}$	0.09908

Table 2: Learned symbolic expressions for Feynman equation $\Delta U = Gm_1m_2(1/r_2 - 1/r_1)$ under noise $\sigma = 0.1$ from VaSST, PySR, Operon, and DSR along with their RMSEs.

Results for learning $P = \frac{\kappa A(T_2 - T_1)}{d}$.

Method	Learned expression	RMSE
VaSST	$P = 0.014782 - 0.805371\left[\frac{AT_1\kappa}{d}\right] + 0.998520\left[\frac{A\kappa T_2}{d}\right] - 0.000254\left[\frac{1}{\sqrt{A}T_1\kappa} - \frac{\frac{d}{\sqrt{A}} - \frac{T_2}{T_1}}{\sqrt{A}T_1\kappa - \frac{T_2}{T_1}}\right]$	0.012602
PySR	$P = \frac{A\kappa(T_2 - T_1)}{d}$	1.59×10^{-16}
Operon	$P = 4.668\frac{\frac{-2.018665\left(\frac{T_2}{T_1}\right)\left(\sqrt{A}T_1\kappa\right) + 2.051\left(\sqrt{A}T_1\kappa\right)}{1.904\sqrt{A}T_1\kappa}}{\sqrt{1+\left[\frac{-4.07664\left(\frac{T_2}{T_1}\right)\left(\sqrt{A}T_1\kappa\right)}{\sqrt{1+\left\{\sin\left(-0.172\frac{T_2}{T_1}\right)+0.516\frac{d}{\sqrt{A}}\right\}^2}}\right]^2}} - 9.425\frac{d}{\sqrt{A}}$	9.66×10^{-5}
DSR	$P = \frac{A\kappa(T_2 - T_1)}{d}$	1.61×10^{-16}

Table 3: Learned symbolic expressions for Feynman equation $P = \kappa A(T_2 - T_1)/d$ in the noiseless setting from VaSST, PySR, Operon, and DSR along with their RMSEs.

Method	Learned expression	RMSE
VaSST	$P = 0.019505 - 0.842846\left[\frac{AT_1\kappa}{d}\right] + 1.005597\left[\frac{A\kappa T_2}{d}\right] - 0.000149\left[\frac{1}{\sqrt{A}T_1\kappa} - \frac{\frac{d}{\sqrt{A}} - \frac{T_2}{T_1}}{\sqrt{A}T_1\kappa - \frac{T_2}{T_1}}\right]$	0.100325
PySR	$P = \frac{A\kappa(T_2 - 1.0040724T_1)}{d}$	0.100212
Operon	$P = 0.538 - 0.678\left[\frac{\exp\left(-0.525\frac{T_2}{T_1}\right) + 1.888\sqrt{A}T_1\kappa - 1.856922\left(\sqrt{A}T_1\kappa\right)\left(\frac{T_2}{T_1}\right)}{\sqrt{1 + \left(1.254\frac{d}{\sqrt{A}}\right)^2}} + \sin\left(\sin\left(\exp\left[\exp\left\{-1.164401\left(\sqrt{A}T_1\kappa\right)\left(\frac{T_2}{T_1}\right)\right\}\right]\right)\right)\right]$	0.099843
DSR	$P =]fracA\kappa(T_2 - T_1 d$	0.09908

Table 4: Learned symbolic expressions for Feynman equation $P = \kappa A(T_2 - T_1)/d$ under noise $\sigma = 0.1$ from VaSST, PySR, Operon, and DSR along with their RMSEs.

Results for learning $f = \beta(1 + \cos\theta)$.

Method	Learned expression	RMSE
VaSST	$f = -0.8994 + 0.00125\cos(2\alpha) + 0.102297\theta + \underline{1.02384\,\beta(1+\alpha\cos\theta)}$	0.13497
PySR	$f = \beta\Big[0.009903\,\alpha(\theta^2 - 24.691)(7.917 - \theta) + 0.29063\Big](10.354 - \theta)$	0.45075
Operon	$f = 0.214 - 3.021\beta\Big[e^{-0.484\theta} + 0.083\alpha + \frac{e^{-0.500\theta}}{0.080\alpha} - \sin(\sin(0.757\theta))\{\sin(0.757\theta) - 0.404\alpha\}\Big]$	0.66695
DSR	$f = \beta(1 + \alpha\cos\theta)$	1.848×10^{-15}

Table 5: Learned symbolic expressions for Feynman equation $f = \beta(1 + \alpha\cos\theta)$ in the noiseless setting from VaSST, PySR, Operon, and DSR along with their RMSEs.

Method	Learned expression	RMSE
VaSST	$f = -0.92928 + 0.000575\cos(2\alpha) + 0.106661\theta + \underline{1.02407\,\beta(1+\alpha\cos\theta)}$	0.16513
PySR	$f = (-2.483891\theta + 20.96369 + \alpha) \times \beta(\theta - 4.86540)(-0.24211\theta + 1.95075)$	1.49573
Operon	$f = 0.377 - 83.187\left[\frac{\cos(-0.981\theta)(-0.368\beta)}{\sqrt{1 + (-1.085\theta)^2}} \times \frac{1}{\sqrt{1 + \left(\frac{\cos(\cos(\cos(-0.999\theta)))}{\sqrt{1+\exp\{\cos(-0.999\theta)\cos(0.340\alpha)\}^2}}\right)^2}}\right](0.318\alpha)$	0.51157
DSR	$f = \beta(1 + \alpha\cos\theta)$	0.09908

Table 6: Learned symbolic expressions for Feynman equation $f = \beta(1+\alpha\cos\theta)$ under noise $\sigma = 0.1$ from VaSST, PySR, Operon, and DSR along with their RMSEs.

Method	Learned expression	RMSE
VaSST	$f = -0.95916 - 0.000097\cos(2\alpha) + 0.110026\theta + \underline{1.02430\,\beta(1+\alpha\cos\theta)}$	0.23748
PySR	$f = \left[-0.02377\left\{\theta + -0.70132(\theta - 9.13492)^2\right\}^2 + 1.22422\right]\alpha\beta$	1.32648
Operon	$f = 3.429 + 0.352\left[\left\{\frac{\cos(0.976\theta)}{\sqrt{1+(0.653\theta)^2}}\right\}(5.549\beta)(2.578\alpha) + \left\{\frac{\cos(-0.781\theta)}{\exp(\cos(1.718\theta))}\right\}(-0.437\alpha) - 1.536\alpha\right]$	0.46428
DSR	$f = \beta(1+\alpha\cos\theta)$	0.1981

Table 7: Learned symbolic expressions for Feynman equation $f = \beta(1+\alpha\cos\theta)$ under noise $\sigma = 0.2$ from VaSST, PySR, Operon, and DSR along with their RMSEs.